

Active fibres and natural boundaries in wild finite-adelic dynamics

Abstract

We give an exact finite criterion for a power series with finite-adelic radial coefficients and a finite sum of unit-modulus exponential phases to be rational or to have the unit circle as a meromorphic natural boundary. The periodic phase weights may be complex, and the periodic valuation and wild exponents may be arbitrary nonnegative real numbers. The criterion first groups phases whose ratios are roots of unity, then tests whether a nonzero grouped weight survives on a residue fibre with unbounded valuation at an active prime. Its proof constructs the actual finite atomic coefficient measure after all coincident frequencies have been combined. An active fibre forces unbounded conductor; invariance under local units then supplies arbitrarily fine grids of nonzero atoms. Radial limits detect these masses and exclude meromorphic continuation. We apply the criterion to confined finite-adelically distorted systems, without a unique dominant root or integral wild exponents, and exhibit an actual nonhyperbolic wild product system. The no-wild application is explicitly separated from the conclusion already obtainable from adelic-perturbation theorems.

1 Introduction

Local distortion of periodic-point counts often changes a rational dynamical zeta function into one with a natural boundary. When several archimedean phases have the same maximal modulus, however, a natural-boundary argument must account for cancellation between them. This issue persists even if every individual phase-weighted summand has a natural boundary: the sum need not have one. We resolve this cancellation question for finite-adelic radial coefficients with periodic data by an exact finite test.

The coefficients under consideration have the form

$$a_n = \left(\sum_{j=1}^J c_j \eta_j^n \right) r_n \prod_{p \in S} p^{-s_{p,n} v_p(n) - t_{p,n} v_p(n)}. \quad (1)$$

Here S and the phase set are finite, $|\eta_j| = 1$, the sequence r_n is periodic and complex, and all $s_{p,n}, t_{p,n}$ are periodic nonnegative real numbers. No algebraicity, height bound, positivity of the phase weights, or independence condition on the phases is imposed. Theorem 2.2 states that $\sum_{n \geq 1} a_n z^n$ is rational precisely when every potentially unbounded active valuation fibre is annihilated by the appropriate grouped phase weight. Otherwise every point of the unit circle obstructs meromorphic continuation.

The test is finite because only the common residue period, the finite phase classes modulo roots of unity, and the primes in S enter it. Its proof is not a no-cancellation principle for arbitrary functions. We first construct an absolutely summable Fourier expansion on the closure of the diagonal integers in a finite residue group times finitely many p -adic integer groups. We then combine all equal frequencies. An active fibre forces unbounded conductor in this actual expansion. Local radial invariance produces full root grids along an unbounded sequence of conductors in the

Result	Relevant hypotheses	Role here
[5, Theorem 11.3.8]	Finite-adelic dynamics; unique dominant root; integral wild data	Classical distorted natural-boundary theorem
[1, Theorem 2.14]	Stable rational series; almost quasi-polynomial perturbation of sublinear height	Already implies the no-wild application
Theorem 2.2	Finite phases and primes; periodic nonnegative real exponents; complex weights	Exact active-fibre test, including phase masking and wild data

Table 1: Different hypothesis classes in the comparison. The theorem locators in the first two rows refer to the public versions specified in the text and bibliography.

nonzero Fourier support, with the same nonzero coefficient throughout each grid. Thus the *actual* atomic measure, rather than a formal decomposition of it, has dense nonzero atoms.

Natural-boundary questions for algebraic dynamics have a substantial history. Bell–Miles–Ward developed a Pólya–Carlson programme for algebraic systems, including finite-place distortion and the role of archimedean unit roots [2]. Byszewski–Cornelissen treated abelian varieties in positive characteristic, with a valuation-kernel dichotomy in the appendix by Royals–Ward [4]. Byszewski–Cornelissen–Houben organized a much wider class through finite-adelically distorted systems [5]. Their inspected Theorem 11.3.8 assumes a unique dominant root and integral wild data. Our dynamical application removes these two assumptions for the finite-adelic presentation considered here.

It is important to distinguish this conclusion from known no-wild results. Baril Boudreau–Holmes–Nguyen prove a natural-boundary alternative for stable rational series perturbed by almost quasi-polynomials [1]. Their theorem already gives the no-wild nonhyperbolic finite-adelic conclusion; we give the deduction in Section 5. The genuinely wild example in Section 6 fails their sublinear-height hypothesis. Likewise, the normalized example does not satisfy the all-embeddings unit-disc convergence condition in the criterion of Bell–Gunn–Nguyen–Saunders [3]. These failed sufficient hypotheses delimit the applications of those theorems; they are not, by themselves, priority assertions.

The Fourier-to-boundary passage is classical in character. For example, Knill–Lesieutre use Fourier expansions and radial singularities for almost-periodic coefficients [7, Proposition 2.2 in the revised manuscript]. Cornelissen–Park also use Fourier analysis of finite-adelic coefficients, including archimedean phases, to study orbit-decomposition statistics [6]. We do not claim a new general Cauchy-transform principle or the first Fourier treatment of these coefficients. The point is the exact active-fibre criterion and the conductor-grid argument proving dense *noncancelled* masses in its active branch.

For clarity about source versions, our detailed comparisons use [5] in its 19 April 2024 public version and [1] in its 16 July 2023 preprint version. The latter’s journal metadata has been verified, but its final text has not been compared line by line with that preprint; no assertion that a later book or final revision is unchanged is needed for the proofs below.

2 Radial data and the finite criterion

Write $\mathbb{T} = \{\xi \in \mathbb{C} : |\xi| = 1\}$, and normalize $v_p(p) = 1$. For $s, t \geq 0$, define a continuous radial function on $\mathbb{Z}_p$ by

$$H_{p;s,t}(x) = \begin{cases} p^{-sv_p(x)-tp^{v_p(x)}}, & x \neq 0, \\ 0, & x = 0, (s,t) \neq (0,0), \\ 1, & x = 0, (s,t) = (0,0). \end{cases} \quad (2)$$

We call the pair (s, t) *active* when it is not $(0, 0)$. For an active pair the values tend to zero as $v_p(x)$ tends to infinity, which explains the value assigned at zero.

Fix the data of (1), with distinct phases $\eta_j \in \mathbb{T}$ and nonzero $c_j \in \mathbb{C}$. Empty phase sums or prime sets are allowed. All periodic sequences are extended to integer indices. Set $u_n = \sum_j c_j \eta_j^n$ and

$$a_n = u_n r_n \prod_{p \in S} H_{p;s_p,n,t_p,n}(n), \quad F(z) = \sum_{n \geq 1} a_n z^n.$$

The coefficients are bounded, so the series defines a holomorphic function in $|z| < 1$.

Partition the phases into equivalence classes by

$$\eta_i \sim \eta_j \iff \eta_i/\eta_j \text{ is a root of unity.}$$

For each class C , choose a representative ρ_C , and write $\eta_j = \rho_C \omega_j$, where ω_j is a root of unity. Choose a positive integer W divisible by all periods of the data and all orders of the ω_j . For $a \in \mathbb{Z}/W\mathbb{Z}$, put

$$b_C(a) = r_a \sum_{j \in C} c_j \omega_j^a, \quad \nu_p = v_p(W). \quad (3)$$

Definition 2.1. The active-fibre condition AF holds if there exist a phase class C , a prime $p \in S$, and a residue $a \in \mathbb{Z}/W\mathbb{Z}$ such that

$$b_C(a) \neq 0, \quad (s_{p,a}, t_{p,a}) \neq (0, 0), \quad a \equiv 0 \pmod{p^{\nu_p}}. \quad (4)$$

When $p \nmid W$, the last condition is vacuous.

Theorem 2.2 (Active-fibre dichotomy). *If AF fails, then F is rational. If AF holds, then the unit circle is a natural boundary for F , and F has no meromorphic continuation through any point of that circle.*

In either case there is a finite complex atomic measure μ on $\mathbb{T}$ with

$$a_n = \int_{\mathbb{T}} \xi^n d\mu(\xi) \quad (n \geq 1). \quad (5)$$

If AF holds, the nonzero atoms of this measure are dense, and for every $\xi_0 \in \mathbb{T}$,

$$\lim_{\varrho \uparrow 1} (1 - \varrho) F(\varrho \xi_0^{-1}) = \mu(\{\xi_0\}). \quad (6)$$

Here a meromorphic continuation through a boundary point means a meromorphic function on a neighborhood of that point agreeing with F on the interior overlap. This local formulation is stronger than excluding one prescribed global continuation.

The criterion does not depend on the representative ρ_C : changing it multiplies $b_C(a)$ by a nonzero root of unity. It is also unchanged when W is replaced by a multiple W' . An active residue modulo W' reduces to one modulo W . Conversely, an active residue modulo W can be lifted to one that is zero modulo $p^{v_p(W')}$, because it is already zero modulo $p^{v_p(W)}$; the remaining prime-power conditions are compatible by the Chinese remainder theorem. The periodic weights and exponents are unchanged on this lift.

Example 2.3 (A completely masked active kernel). Let $S = \{2\}$, $r_n = 1$, $s_{2,n} = 1$, $t_{2,n} = 0$, and $u_n = 1 - (-1)^n$. The phase sum vanishes on the even integers, whereas the radial factor is one on the odd integers. Hence

$$F(z) = \sum_{n \geq 1} (1 - (-1)^n) 2^{-v_2(n)} z^n = \frac{2z}{1 - z^2}.$$

There is one phase class and $W = 2$; its grouped weight vanishes on the only residue with unbounded 2-adic valuation. Thus AF fails. Treating the two phases separately would miss this cancellation.

3 An absolutely summable adelic Fourier representation

The compact group that keeps the periodic and local coordinates compatible is

$$D = \overline{\{(n \bmod W, (n)_{p \in S}) : n \in \mathbb{Z}\}} \subset (\mathbb{Z}/W\mathbb{Z}) \times \prod_{p \in S} \mathbb{Z}_p.$$

Let $g_0 = (1 \bmod W, (1)_{p \in S})$. The Chinese remainder theorem gives the explicit description

$$D = \{(a, x) : x_p \equiv a \pmod{p^{\nu_p}} \text{ for every } p \in S\}. \quad (7)$$

Indeed, the right-hand side is closed and contains every diagonal integer. Conversely, a neighborhood of a point on the right specifies the finite coordinate modulo W and each local coordinate modulo some p^{K_p} , with $K_p \geq \nu_p$. Compatibility in (7) is precisely the condition for this simultaneous congruence system to have an integer solution.

For any continuous characters χ, χ' of D ,

$$\chi(g_0) = \chi'(g_0) \implies \chi = \chi', \quad (8)$$

since equality then holds on all integer multiples of g_0 , a dense subgroup. All characters used below are restrictions of finite-order product characters, so their values at g_0 are roots of unity.

Lemma 3.1 (Radial Fourier expansion). *Each function $H_{p;s,t}$ has a uniformly absolutely convergent character expansion on $\mathbb{Z}_p$, with sum of absolute coefficients at most 2. For each phase class C , the function*

$$q_C(a, x) = b_C(a) \prod_{p \in S} H_{p;s_p,a,t_p,a}(x_p) \quad (9)$$

therefore has an expansion on D

$$q_C(g) = \sum_{\chi} \hat{q}_C(\chi) \chi(g), \quad \sum_{\chi} |\hat{q}_C(\chi)| \leq 2^{|S|} \sum_{a \in \mathbb{Z}/W\mathbb{Z}} |b_C(a)|. \quad (10)$$

The coefficients in this expression are the Haar Fourier coefficients after equal restricted characters have been combined.

Proof. For an active pair, put $h(v) = p^{-sv - tp^v}$ for $v \geq 0$ and $\delta_j = h(j-1) - h(j)$. The sequence h decreases strictly to zero, so $\delta_j > 0$ and $\sum_{j \geq 1} \delta_j = h(0) \leq 1$. Telescoping yields

$$H_{p;s,t}(x) = h(0) - \sum_{j \geq 1} \delta_j \mathbf{1}_{p^j \mathbb{Z}_p}(x). \quad (11)$$

At $x \neq 0$, the sum telescopes to $h(v_p(x))$; at zero it telescopes to zero. Its functional tail is bounded uniformly by its numerical tail.

Writing $[x]_{p^j}$ for a residue representative, the finite orthogonality identity is

$$\mathbf{1}_{p^j \mathbb{Z}_p}(x) = \frac{1}{p^j} \sum_{b=0}^{p^j-1} \exp\left(\frac{2\pi i b [x]_{p^j}}{p^j}\right). \quad (12)$$

Thus each ball indicator has Fourier norm one. Combining (11) and (12) gives total absolute coefficient mass at most $2h(0) \leq 2$. The inactive kernel is simply the trivial character.

On the ambient product group, express (9) as a finite sum over the singleton indicators of $a \in \mathbb{Z}/W\mathbb{Z}$. Each such indicator also has Fourier norm one. Multiplication of the absolutely summable local expansions, followed by this finite sum, gives the bound in (10). Restricting to D and combining characters that coincide cannot increase the sum of absolute coefficients.

Finally, uniform absolute convergence permits termwise integration against a conjugate character. A nontrivial character has Haar integral zero: translation by a point at which it differs from one multiplies the integral by that value while leaving it invariant. This identifies the combined coefficients with the Fourier integrals. $\square$

The lemma is a consequence of the particular monotone radial kernels. No absolute-Fourier-convergence assertion for arbitrary continuous functions on a compact group is being used.

4 Conductor grids and the exact dichotomy

Lemma 4.1 (An active fibre has unbounded conductor). *Suppose C, p, a satisfy (4). In the expansion of q_C there are nonzero coefficients with arbitrarily high p -conductor. More precisely, for every $K \geq \nu_p$, some character with nonzero coefficient is nontrivial on the subgroup having finite coordinate zero, p -coordinate in $p^K \mathbb{Z}_p$, and all other coordinates zero.*

Proof. Fix the finite coordinate at a . For every $q \neq p$, choose a nonzero element x_q in $a + q^{\nu_q} \mathbb{Z}_q$. Every corresponding kernel value is strictly positive. Along the remaining coordinate ball $p^{\nu_p} \mathbb{Z}_p$, formula (9) becomes

$$q_C(a, x) = B_0 H_{p; s_p, a, t_p, a}(x_p), \quad B_0 \neq 0.$$

Its values at $x_p = 0$ and $x_p = p^K$ differ for every $K \geq \nu_p$. If all characters with nonzero coefficient were trivial on the subgroup in the statement, uniform convergence in (10) would make q_C invariant under its translations, contradicting these two values.

Represent such a character using a product character whose p -component has the form

$$x_p \mapsto \exp(2\pi i b [x_p]_{p^k} / p^k), \quad p \nmid b.$$

Nontriviality on $p^K \mathbb{Z}_p$ forces $k > K$. When $k > \nu_p$, this high conductor cannot be absorbed in the finite coordinate: its restriction to the p -coordinate subgroup $p^{\nu_p} \mathbb{Z}_p$ still has conductor $p^{k-\nu_p}$. Thus these conductors are unbounded in one fixed class C and at one fixed prime p . $\square$

Lemma 4.2 (Full grids of actual nonzero coefficients). *Under the hypotheses of Lemma 4.1, the nonzero frequencies $\chi(g_0)$ of q_C contain rotated full root grids of unbounded orders, with a common nonzero Fourier coefficient on each grid. In particular, those frequencies are dense in $\mathbb{T}$.*

Proof. Set $\kappa = \max(1, \nu_p)$. For a p -adic unit $v \equiv 1 \pmod{p^\kappa}$, let α_v multiply x_p by v and leave every other coordinate, including a , fixed. Equation (7) shows that α_v is an automorphism of D . Radiality implies $q_C \circ \alpha_v = q_C$. Invariance of Haar probability under this automorphism gives

$$\widehat{q}_C(\chi \circ \alpha_v) = \widehat{q}_C(\chi). \quad (13)$$

Take a nonzero coefficient with p -conductor p^k , $k > \kappa$, and primitive numerator b , as in Lemma 4.1. For $v = 1 + p^\kappa t$, $0 \leq t < p^{k-\kappa}$, evaluation at g_0 gives

$$(\chi \circ \alpha_v)(g_0) = \chi(g_0) \exp\left(\frac{2\pi i b t}{p^{k-\kappa}}\right). \quad (14)$$

Since $p \nmid b$, these are the distinct points of a rotated full grid of order $p^{k-\kappa}$. Equation (13) assigns the same nonzero coefficient to each point. The conductors k are unbounded, so every nonempty open arc contains a point of one of these grids. $\square$

Unbounded conductor alone would not suffice: an arbitrary infinite set of rational frequencies need not be dense. The complete unit orbits in (14) are the step that supplies density.

Proof of Theorem 2.2. Evaluation on the diagonal integers and undoing the phase grouping give

$$a_n = \sum_C \rho_C^n q_C(n g_0) = \sum_C \sum_\chi \widehat{q}_C(\chi) (\rho_C \chi(g_0))^n.$$

Define the finite-variation atomic measure

$$\mu = \sum_C \sum_\chi \widehat{q}_C(\chi) \delta_{\rho_C \chi(g_0)}. \quad (15)$$

Its total variation is finite by Lemma 3.1. Within one class, distinct characters have distinct generator values by (8). Between different classes, an equality $\rho_C \chi(g_0) = \rho_{C'} \chi'(g_0)$ would make $\rho_C / \rho_{C'}$ a root of unity, contradicting the definition of the classes. Thus no further collisions remain in (15). Each displayed nonzero coefficient is a nonzero mass of the actual measure. Equation (5) follows, and if AF holds, Lemma 4.2 proves density of its nonzero atoms.

For $|z| < 1$, absolute summability allows the geometric series to be integrated term by term:

$$F(z) = \int_{\mathbb{T}} \frac{\xi z}{1 - \xi z} d\mu(\xi). \quad (16)$$

For $\xi, \xi_0 \in \mathbb{T}$ and $0 < \varrho < 1$,

$$\left| (1 - \varrho) \frac{\varrho \xi \xi_0^{-1}}{1 - \varrho \xi \xi_0^{-1}} \right| \leq \varrho \leq 1.$$

The expression tends to one if $\xi = \xi_0$ and to zero otherwise. Dominated convergence with respect to $|\mu|$ proves (6), including at points that are not atoms.

A nonzero right-hand side in (6) excludes a holomorphic extension at ξ_0^{-1} , since local boundedness would force that limit to vanish. The inverse atom points are dense, so no boundary arc admits holomorphic continuation. If a meromorphic continuation existed in a neighborhood of any boundary point, every inverse nonzero atom in the enclosed arc would have to be a pole. Poles of a meromorphic function are discrete, which is incompatible with this dense set. This proves the active branch.

Suppose instead that AF fails. On a fibre over a , either $b_C(a) = 0$, or each active coordinate has $a \not\equiv 0 \pmod{p^{\nu_p}}$. In the latter situation its valuation is fixed throughout $a + p^{\nu_p} \mathbb{Z}_p$, strictly below

ν_p . Every inactive factor equals one. Hence q_C is constant on the fibre; denote this value by $d_C(a)$. With the zero residue represented by W , summation on residue classes yields

$$F(z) = \sum_C \frac{\sum_{a=1}^W d_C(a)(\rho_C z)^a}{1 - (\rho_C z)^W}. \quad (17)$$

This proves rationality, including the empty-data cases. $\square$

5 Finite-adelically distorted dynamical systems

A self-map $f : X \rightarrow X$ is *confined* if $f_n = \# \text{Fix}(f^n)$ is finite for every $n \geq 1$. Its fixed-count series and dynamical zeta function are

$$Z_f(z) = \sum_{n \geq 1} f_n z^n, \quad \zeta_f(z) = \exp \left(\sum_{n \geq 1} \frac{f_n}{n} z^n \right).$$

A sequence of period m is a *gcd sequence* if its value at n depends only on $\gcd(n, m)$. We use the following finite-adelically distorted, or FAD, presentation from [5, Definitions 7.1.1–7.1.2]:

$$f_n = |\det(A^n - I)| c^n r_n \prod_{p \in S} |n|_p^{s_{p,n}} p^{-t_{p,n} |n|_p^{-1}}. \quad (18)$$

Here A is an integral square matrix, $c > 0$, S is finite, r_n is a positive real gcd sequence, and $s_{p,n}, t_{p,n}$ are nonnegative real gcd sequences having periods coprime to their own prime p . A prime is *genuinely distorted* if its exponent pair is active at some residue. An exponent $t_{p,n} \neq 0$ is the wild part of the distortion.

We use counting entropy, namely $\limsup_{n \rightarrow \infty} n^{-1} \log f_n = \log \Lambda$ for a nonzero count sequence. The standard FAD entropy formula identifies $\Lambda = c \prod_{|\lambda| > 1} |\lambda|$, with the eigenvalues of A counted with multiplicity [5, Section 10.3]. Our argument below also identifies the radius Λ^{-1} directly whenever distortion is present.

Corollary 5.1. *Let a nonzero confined FAD system have positive counting entropy $\log \Lambda > 0$. If at least one prime in (18) is genuinely distorted, then both Z_f and ζ_f have $|z| = \Lambda^{-1}$ as a natural boundary, with no meromorphic continuation through any point of that circle. No unique-dominant-root hypothesis or integrality assumption on the wild exponents is required.*

Proof. First, A has no root-of-unity eigenvalue. If such an eigenvalue had order h , then $f_n = 0$ at every multiple of h . A positive f_m would supply a point fixed also by f^{hm} , a contradiction. The assumed nonzero count sequence therefore excludes this possibility.

The determinant factor has an expansion

$$|\det(A^n - I)| c^n = \Lambda^n u_n + O((\Lambda \tau)^n), \quad 0 \leq \tau < 1, \quad (19)$$

where u_n is a finite sum of unit phases and is strictly positive for every positive integer. This is the classical dominant expansion in [5, Section 10.3, especially Lemma 10.3.10]. Its relevant features can also be seen directly. After the moduli of the eigenvalues outside the unit circle have been extracted, the off-circle factors tend to one exponentially. Unit-circle eigenvalues occur in nonreal conjugate pairs, since the real possibilities 1 and -1 have been excluded. Their product contributes

$$u_n = \prod_{\{\eta, \bar{\eta}\}} (2 - \eta^n - \eta^{-n}) > 0.$$

Expanding this finite product gives the required finite phase sum; the empty product is one.

Let p be a genuinely distorted prime, and choose an active exponent residue $n_0 \bmod m_p$, where m_p is a common period of its two exponent sequences and $p \nmid m_p$. Choose W for the dominant coefficient series as in Section 2. Since $m_p \mid W$, the Chinese remainder theorem supplies $a \bmod W$ satisfying

$$a \equiv 0 \pmod{p^{v_p(W)}}, \quad a \equiv n_0 \pmod{m_p}.$$

The exponent pair is active at a . If $b_C(a) = 0$ for every phase class, positivity of r_a would make every grouped periodic phase sum vanish there. Consequently u_n would vanish for all positive $n \equiv a \pmod{W}$, contradicting $u_n > 0$. Thus AF holds.

Apply Theorem 2.2 to the normalized leading part of (19), multiplied by the periodic and local factors. It has dense nonzero radial masses on the unit circle. Those factors are bounded: the local kernels are at most one and r_n has a finite period. Hence the normalized error in (19) generates a function holomorphic in a disc of radius greater than one. Adding it does not change any nonzero radial mass or remove the resulting dense singularities. Rescaling proves the assertion for Z_f .

Within $|z| < \Lambda^{-1}$, the defining exponential for ζ_f converges and is nonvanishing, and

$$z \frac{\zeta'_f(z)}{\zeta_f(z)} = Z_f(z). \quad (20)$$

If ζ_f had a meromorphic continuation through a boundary point, it would not be identically zero in the continued neighborhood, because it agrees with the nonvanishing exponential on the interior overlap. Its zeros and poles would be discrete. A smaller nonempty boundary arc would therefore avoid them, and (20) would continue Z_f holomorphically through that arc. This contradicts the dense singularities already obtained. $\square$

The own-prime coprimality of the exponent periods has one precise function in this argument: it permits an active exponent value and an unbounded valuation fibre to occur together. Positivity of the leading factor then prevents all phase classes from being masked. Neither hypothesis is needed in the abstract criterion, which retains the rational masked cases.

5.1 The already-known no-wild deduction

We record why removing unique dominance only in the no-wild FAD case is not an additional contribution. Suppose every $t_{p,n} = 0$, and write $f_n = A_n g(n)$, where

$$A_n = |\det(A^n - I)|c^n, \quad g(n) = r_n \prod_{p \in S} p^{-s_{p,n} v_p(n)}.$$

For an actual FAD system, [5, Proposition 10.2.1] gives $c \in \mathbb{Z}_{>0}$. Since the wild exponents are zero and hence integral, [5, Proposition 10.2.2] gives rationality of r_n and $p^{s_{p,n}}$. Therefore $g(n) \in \mathbb{Q}$, and its logarithmic Weil height satisfies $h(g(n)) = O(\log n)$.

If S is empty, g is periodic and already an almost quasi-polynomial. Otherwise, let W be a common period and put $d_k = W \prod_{p \in S} p^k$. On each residue class modulo d_k which avoids zero modulo p^k for every $p \in S$, all relevant valuations and periodic data are fixed. Thus g is constant there. The proportion of excluded classes is at most $\sum_{p \in S} p^{-k}$, tending to zero. These rationality, height, and density-one local polynomial conditions make g an almost quasi-polynomial in the sense of [1, Definition 2.11], with constant constituent polynomials.

The generating function of A_n is rational: expansion of the determinant gives finitely many exponential terms, and its real sign is periodic. Its dominant part is strictly positive by (19), so it

vanishes identically on no arithmetic progression. It is therefore stable in the sense required by [1, Theorem 2.14]. That theorem yields the rational or natural-boundary alternative for $\sum f_n z^n$. If a valuation exponent is active, the FAD rationality criterion [5, Proposition 11.3.3] excludes rationality. The logarithmic derivative then transfers the boundary to ζ_f . In particular, the abelian-variety and finite-place solenoid applications already settled in [1] are not being asserted anew here.

The classical root-rational classification for FAD zeta functions [5, Theorem 11.3.4] is a separate input and is not reproved as part of Theorem 2.2. Our additional analytic argument is needed for wild data beyond the preceding height hypothesis, as the next example makes explicit.

6 A realized nonhyperbolic wild example

Consider the polynomial and its companion matrix

$$P(X) = X^4 - X^3 - X^2 - X + 1, \quad A = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Since $\det A = 1$, this matrix induces a toral automorphism $T_A : \mathbb{R}^4/\mathbb{Z}^4 \rightarrow \mathbb{R}^4/\mathbb{Z}^4$. Modulo two, the polynomial becomes $X^4 + X^3 + X^2 + X + 1$. It has no root in $\mathbb{F}_2$, and reduction modulo the only monic irreducible quadratic $X^2 + X + 1$ gives $X + 1$, not zero. It is therefore irreducible over $\mathbb{F}_2$, and Gauss's lemma implies irreducibility over $\mathbb{Q}$.

For $X \neq 0$, put $Y = X + X^{-1}$. Then

$$X^{-2}P(X) = Y^2 - Y - 3.$$

The two Y -roots are $(1 + \sqrt{13})/2 > 2$ and $(1 - \sqrt{13})/2 \in (-2, 2)$. Consequently the roots of P are $\lambda, \lambda^{-1}, \eta, \eta^{-1}$, where $\lambda > 1$, $|\eta| = 1$, and η is nonreal. None is a root of unity: irreducibility makes the real root $\lambda > 1$ a conjugate of η , whereas every conjugate of a root of unity has modulus one. Thus $A^n - I$ is nonsingular for all $n \geq 1$. Its integer lattice index, or Smith normal form, gives

$$\# \text{Fix}(T_A^n) = |\det(A^n - I)| = (\lambda^n + \lambda^{-n} - 2)(2 - \eta^n - \eta^{-n}) > 0. \quad (21)$$

Now fix an odd prime p and the additive-group endomorphism

$$U : \overline{\mathbb{F}}_p \rightarrow \overline{\mathbb{F}}_p, \quad U(x) = x^p + x.$$

Let $\Phi(x) = x^p$, so $U = 1 + \Phi$. If $n = p^a m$, $p \nmid m$, then in $\mathbb{F}_p[T]$

$$(1 + T)^n - 1 = ((1 + T)^m - 1)^{p^a}.$$

Its lowest nonzero T -degree is p^a , and its highest is n . Substituting Φ , the additive polynomial $U^n - 1$ has ordinary degree p^n , with smallest nonzero monomial of degree p^{p^a} . Over the perfect field $\overline{\mathbb{F}}_p$, it is a p^{p^a} -th power of a separable polynomial of degree p^{n-p^a} and nonzero linear coefficient. That separable polynomial splits with distinct roots. Hence

$$\# \text{Fix}(U^n) = p^{n-p^{p^a}}. \quad (22)$$

Take the actual product self-map

$$f = T_A \times U \quad \text{on} \quad X = (\mathbb{R}^4/\mathbb{Z}^4) \times \overline{\mathbb{F}}_p.$$

Equations (21)–(22) give

$$f_n = |\det(A^n - I)| p^n p^{-p^{v_p(n)}}. \quad (23)$$

All fixed counts are positive and finite. This is a FAD presentation with $c = p$, $r_n = 1$, $S = \{p\}$, $s_{p,n} = 0$, and $t_{p,n} = 1$. Every periodic datum has period one.

Put $\Lambda = p\lambda$. The leading normalized factor is $2 - \eta^n - \eta^{-n}$, so there are three dominant characteristic roots

$$\Lambda, \quad \Lambda\eta, \quad \Lambda\eta^{-1},$$

with nonzero coefficients 2, -1 , -1 , respectively. They belong to different root-of-unity ratio classes. Indeed, torsion of η^2 would imply torsion of η . Thus the example is not hyperbolic in the unique-dominant-root sense of [5, Definition 10.3.9], and no iterate removes this issue. Corollary 5.1 gives the meromorphic natural boundary

$$|z| = (p\lambda)^{-1} \quad (24)$$

for both its fixed-count series and its dynamical zeta function. The entropy here is counting entropy. No topological entropy identity for this mixed product set is being assumed.

6.1 Two sufficient criteria that do not cover this normalization

The normalized distortion is $g(n) = p^{-p^{v_p(n)}}$. At $n = p^a$, its absolute logarithmic Weil height is

$$h(g(p^a)) = p^a \log p = n \log p.$$

Thus it fails the condition $h(g(n)) = o(n)$ in the inspected almost-quasi-polynomial theorem [1, Definition 2.11]. Local constancy on a density tending to one of residue classes does not repair this height failure. Moving p^n into the perturbation also fails: on integers coprime to p , the resulting factor has height $(n - 1) \log p$.

There is a separate obstruction to applying the all-embeddings criterion in [3, Theorems 1.2 and 1.6] to the normalized coefficients $a_n = f_n / (p\lambda)^n$. An embedding of $\mathbb{Q}(\lambda)$ sending λ to λ^{-1} exists by irreducibility. Because f_n is an integer, it sends a_n to $\lambda^{2n} a_n$. On the progression $n \equiv 1 \pmod{p}$, the wild factor is p^{-1} , and the irrational rotation η^{1+p^k} has a subsequence for which $2 - \eta^n - \eta^{-n}$ is bounded below by a positive constant. It follows from (21) that $\limsup |a_n|^{1/n} = 1$, while the conjugated normalized series has radius $\lambda^{-2} < 1$. Unit-disc convergence at every embedding is therefore unavailable.

The factors of this example and their fixed-count formulas are classical ingredients; their derivations are included to verify realization and the removed hypothesis. The product is a self-map of an actual set, not an asserted algebraic group spanning two characteristics.

7 Conclusions and scope

For the coefficient class (1), rationality is equivalent to the disappearance of every active unbounded valuation fibre after torsion-related phases have been aggregated. An unmasked fibre forces dense actual Fourier masses and, by a radial limit, a natural boundary even for meromorphic continuation. This gives a finite exact criterion, not merely a sufficient generic noncancellation condition.

The assumptions are part of the conclusion's scope: only finitely many primes and phases are allowed; the weights and exponents are periodic; and the exponents are nonnegative real numbers. The proof does not extend by assertion to infinitely many distortion primes, arbitrary continuous local functions, or general sums of natural-boundary series.

For realized FAD systems, the own-prime coprimality of exponent periods and positivity of the determinant’s leading factor force the active branch whenever genuine distortion is present. The no-wild application is already obtained from the cited adelic-perturbation theorem. The additional wild application includes the explicit nonhyperbolic product in Section 6, without a height or all-embeddings assumption. The statements concern ordinary integer iterates and their native fixed-point counts; no change of clock or identification with a different arithmetic zeta function is made.

References

- [1] Félix Baril Boudreau, Erik Holmes, and Khoa D. Nguyen. Adelic perturbation of rational functions and applications. *Mathematische Annalen*, 392(2):2253–2275, 2025. doi: 10.1007/s00208-025-03155-0. URL <https://arxiv.org/abs/2307.07910v1>. Theorem locators refer to arXiv:2307.07910v1, 16 July 2023.
- [2] Jason Bell, Richard Miles, and Thomas Ward. Towards a Pólya–Carlson dichotomy for algebraic dynamics. *Indagationes Mathematicae*, 25(4):652–668, 2014. doi: 10.1016/j.indag.2014.04.005. URL <https://shura.shu.ac.uk/17223/>.
- [3] Jason P. Bell, Keira Gunn, Khoa D. Nguyen, and J. C. Saunders. A general criterion for the Pólya–Carlson dichotomy and application. *Transactions of the American Mathematical Society*, 376(6):4361–4382, 2023. doi: 10.1090/tran/8876. URL <https://arxiv.org/abs/2206.00862v1>.
- [4] Jakub Byszewski and Gunther Cornelissen. Dynamics on abelian varieties in positive characteristic. *Algebra & Number Theory*, 12(9):2185–2235, 2018. doi: 10.2140/ant.2018.12.2185. URL <https://msp.org/ant/2018/12-9/p06.xhtml>. With an appendix by Robert Royals and Thomas Ward.
- [5] Jakub Byszewski, Gunther Cornelissen, and Marc Houben. Dynamics of endomorphisms of algebraic groups. arXiv:2209.00085v2, 2024. URL <https://arxiv.org/abs/2209.00085v2>. Public version 2, 19 April 2024; first posted in 2022.
- [6] Gunther Cornelissen and Sun Woo Park. Orbit decomposition statistics for discrete dynamical systems: the Cesàro mean and a large deviation principle. arXiv:2605.24504v2, 2026. URL <https://arxiv.org/abs/2605.24504v2>. Version 2, 19 June 2026.
- [7] Oliver Knill and John Lesieutre. Analytic continuation of Dirichlet series with almost periodic coefficients. *Complex Analysis and Operator Theory*, 6(1):237–255, 2012. doi: 10.1007/s11785-010-0064-7. URL https://people.math.harvard.edu/~knill/kam/papers/denjoy_revised.pdf. Proposition locator refers to the author-hosted revised manuscript.